The Attention Span of the World

Measuring and Comparing Media-Attention Lifespans Across Global Event Categories

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The Challenge

Problem: Some major events dominate global news for days or weeks, while others disappear quickly.

Importance: Understanding these attention patterns can show how information spreads, which stories sustain public interest, and which events quickly lose visibility in the global news cycle.

Community: Media researchers, communication scholars, journalists, investors, corporate analysts, policymakers, and researchers studying information diffusion.

Research Question: How do the attention lifespans of technology announcements, scientific discoveries, corporate events, and politics differ in global news coverage?

Potential Impact: This project will use **GDELT** to measure and compare attention curves, including peak coverage, decline rate, geographic spread, and attention half-life across different event categories.

Method: Unsupervised clustering (HDBSCAN/K-Means) will be used to discover natural event groups and recurring attention-lifecycle patterns from GDELT themes, entities, and temporal coverage features.

The Data — Show Us the Actual Data

Dataset: What is the dataset and who is the authoritative source?

Dataset is **GDELT 2.0 Global Knowledge Graph (GKG)**, maintained by the GDELT Project. GKG processes worldwide news and extracts themes, people, organizations, locations, tone, images, quotations, and other metadata. It updates **every 15 minutes.**

Access: How did you actually obtain or access it?

Downloaded directly from GDELT's raw-data master file using gdeltloader. For our initial sample, we downloaded the **20 most recent GKG intervals**.The files were then loaded into a local MongoDB gdelt.gkg collection using Python.

The Data — Show Us the Actual Data

Scale: How large is it in GB? Approximately how many records, files, and columns?

-> 20 fifteen-minute GKG files covering ~5 hours of global news, with:

- 33,614 records
- 27 field
- ~564 MB of raw data.

This means, 564 MB/5 hours≈112.8 MB/hour

1 day ≈ 2.7 GB

Estimated monthly scale: ~80 GB at the observed ingestion rate.

The Data — Show Us the Actual Data

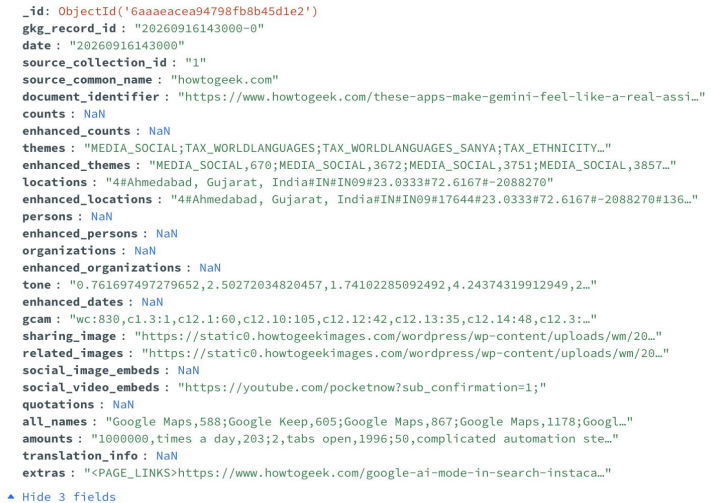
Format: CSV, JSON, Parquet, database, API, etc. What does the schema look like?

Raw files are **tab-separated GKG files** (.gkg.csv) and are stored as MongoDB documents for analysis. Key fields include:

- date
- source
- URL
- themes
- organizations
- persons
- locations
- tone
- images
- quotations

The Data — Show Us the Actual Data

Show Us the Data: Include an actual sample—a table, screenshot, JSON fragment, schema, or other concrete representation.





The Data — Show Us the Actual Data

What Have You Found Already? Show one or two things you noticed about quality, missingness, messiness, structure, or other challenges.

Few things we have noticed in the data:

- **High Null Values:** Entity fields such as persons, organizations, locations, and quotations are frequently missing because not every article contains every entity type.

- **No predefined event-category column:** “Technology announcement,” “scientific discovery,” and “corporate event” are **not predefined labels**; they must be constructed from GDELT themes and entities. This becomes an important data-preparation step.

- **Complex nested fields:** Variables such as themes, enhanced_themes, locations, tone, and GCAM contain multiple values packed into a single field and will require parsing before EDA.

Why This Is a Good EAS 587 Project

Data Intensive: GDELT produces large volumes of time-partitioned datas. Our analysis requires processing many files and connecting event records with repeated media mentions over time.

Scale Challenge: Full data is too large and continuously growing to repeatedly load and analyze in a single pandas DataFrame.

Initial Strategy: Selected event categorizes and event-centered time windows. Process files in batches and store relevant Events and Mentions in MongoDB for indexed retrieval and analysis

Biggest Risk: Single real-world event may correspond to multiple GDELT records and our event categorizes are not directly available as simple GDELT labels. Correctly identifying and grouping event is therefore a major challenge.

Next Step: Generate the first media-attention timeline.

The Pipeline:

```
GDELT 2.0 → Acquire Raw Files → Select Event Windows → MongoDB → Process Events + Mentions
→ Build Attention Curves → Compare Event Categories
```